








Time Series Analysis

Seminar 9: Machine learning for time series



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Acronyms used in this material

- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **AUC**: Area Under the (ROC) Curve
- **BET**: Bucharest Exchange Trading (index)
- **CV**: Cross-Validation
- **DHR**: Dynamic Harmonic Regression
- **DM**: Diebold–Mariano (test)
- **ENTSO-E**: European Network of Transmission System Operators for Electricity
- **EODHD**: EOD Historical Data (market-data provider)
- **EU**: European Union
- **GB**: Gradient Boosting (ensemble of regression trees)
- **HICP**: Harmonised Index of Consumer Prices
- **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- **LSTM**: Long Short-Term Memory (network)
- **MAE**: Mean Absolute Error
- **MASE**: Mean Absolute Scaled Error
- **MIMO**: Multiple-Input Multiple-Output (forecasting strategy)
- **ML**: Machine Learning
- **OLS**: Ordinary Least Squares
- **RF**: Random Forest
- **RMSE**: Root Mean Squared Error
- **ROC**: Receiver Operating Characteristic (curve)
- **RW**: Random Walk (forecast)
- **SSE**: Sum of Squared Errors

Today's Question and Seminar Structure

- The question of the seminar
  - **how do we turn a time series into a machine-learning problem, and how do we know whether the result beats a simple benchmark?**
- Results to obtain (practice only: nothing is handed in or graded)
  - the answers to the A exercises
  - a figure and a table for each B exercise
  - the corrected AI answer of C2
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- The exercises, their types and their order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	a lag table, recursive and direct forecasts; leakage in a rolling mean	Solved, Proposed	A1
A3, A4	one split of a regression tree; ridge and lasso in one dimension	Solved, Proposed	A3
A5, A6	a conformal interval and the pinball loss; MASE and a DM statistic	Solved, Proposed	A5
B1, B2	electricity load: boosting against the seasonal naive; recursive, direct, holidays	Solved, Proposed	B1
B3, B4	Romanian inflation; the sign of BET returns	Solved, Proposed	B3
C1, C2	a global model as a project; what is wrong in an AI answer?	Proposed	B1, B3

- Types: **A** calculations on paper; **B** real data, in the notebook; **C** open questions, an AI critique
- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in the seminar

Data used

Series	Source	Frequency	Period
Electricity load, Romania	ENTSO-E	hourly, averaged per day (GW)	2022–2026
HICP, 27 EU countries	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	2005–2026
BET	EODHD	daily, close	2000–2026

- ENTSO-E: the European Network of Transmission System Operators for Electricity; the same load file as in Chapter 4
- In the notebook: `direct_frame`, `forecast_direct`, `forecast_recursive`, `dm_raw`, `inflation_forecasts`, `sign_frame`; scikit-learn models with fixed seeds; no account or key is needed

Prerequisites for Today (1/17): from a series to a table

- **Supervised learning**: learn a function $\hat{f}$ from pairs (features $\mathbf{x}_i$, target y_i) so that $\hat{f}(\mathbf{x})$ is close to y on **new** data
  - $\mathbf{x}_i$: the vector of features of row i; the model never sees the targets of the test data
- For a series $y_1, \ldots, y_n$: one row per **forecast origin** t

$$y_{t+h} = f(\mathbf{x}_t) + e_{t+h}, \qquad \mathbf{x}_t = (y_t,\ y_{t-1}, \ldots, y_{t-p+1},\ \text{rolling means},\ \text{calendar})$$

  - h: the horizon; p: the number of lags; e_{t+h}: the error; every feature must be known at time t
  - a rolling mean of the last w values: $\bar{y}_t^{(w)} = \frac{1}{w} \sum_{j=0}^{w-1} y_{t-j}$; calendar: weekday, month, holidays of the target day
- **Number of rows**: a series of length n with p lags gives $n - p - h + 1$ complete rows for horizon h
  - the first $p - 1$ origins lack lags; the last h origins lack the target

Prerequisites for Today (2/17): one row of the table in a chart



- Green: the four lags known at the origin $t = 24$, the features $\mathbf{x}_t$; red: the target y_{t+3}
- Moving the origin one step to the right gives the next row: the table is a sliding window over the series

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Prerequisites for Today (3/17): multi-step strategies

- **Recursive**: one model for $h = 1$, applied again with its own forecasts as inputs Ben Taieb et al. (2012)

$$\hat{y}_{t+1} = \hat{f}(y_t, y_{t-1}, \dots), \qquad \hat{y}_{t+2} = \hat{f}(\hat{y}_{t+1}, y_t, \dots)$$

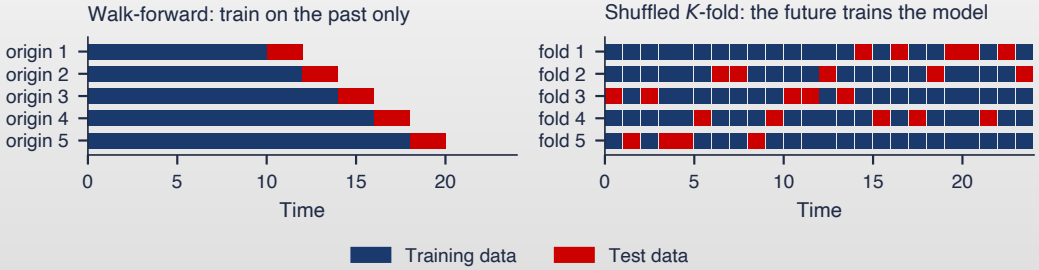
  - one model only; but the errors of the first steps enter the next ones
- **Direct**: a separate model $\hat{f}_h$ for each horizon, trained on the target y_{t+h}

$$\hat{y}_{t+h} = \hat{f}_h(y_t, y_{t-1}, \dots), \qquad h = 1, \dots, H$$

  - H: the longest horizon; no accumulation of errors, but H models to train
- **MIMO** (multiple-input multiple-output): one model with H outputs $(\hat{y}_{t+1}, \dots, \hat{y}_{t+H})$

Prerequisites for Today (4/17): validation over time

- **Walk-forward** validation: train on the data up to the origin, forecast the next block, move the origin
  - the errors are pooled over all origins $t_1 < t_2 < \cdots < t_N$: each forecast uses only the past, as in real time
- K-**fold** cross-validation: split the rows at random in K groups, test on each group in turn
  - for a series, future rows end up in the training data; with overlapping targets the test rows almost repeat training rows
  - never shuffled K-fold for forecasting models with overlapping targets or exogenous features



Prerequisites for Today (5/17): leakage

- **Leakage**: information that is not available at the origin enters the features or the training Kaufman et al. (2012)
  - the test error then looks better than any real forecast can be
- **Rolling mean** feature for the target y_t: only values before t

$$m_t = \frac{1}{w} \sum_{j=1}^{w} y_{t-j} \qquad \text{(in pandas: } \texttt{rolling(w).mean().shift(1)}\text{)}$$

  - without the shift, the window ends at y_t and the feature contains the target
- Other typical leaks
  - scaling (means, standard deviations) estimated on the full sample, before the split
  - the actual weather of the target day instead of the forecast available at the origin
  - calendar variables (weekday, holidays) are known in advance: no leak

Prerequisites for Today (6/17): benchmarks and accuracy measures

- **Benchmarks** (Chapter 0)

$$\text{naive: } \hat{y}_{T+h} = y_T, \qquad \text{seasonal naive: } \hat{y}_{T+h} = y_{T+h-m}$$

  - T: the origin (last observation); m: the seasonal period ($m = 7$ for daily data, for $h \leq m$: the same weekday last week)

- **Errors** $e_i = y_i - \hat{y}_i$ over N test forecasts

$$\text{MAE} = \frac{1}{N}\sum_{i=\mathbf{1}}^{N} |e_i|, \qquad \text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=\mathbf{1}}^{N} e_i^{\mathbf{2}}}$$

  - MAE: mean absolute error, in the units of y; RMSE: root mean squared error, penalises large errors more

- **MASE** Hyndman and Koehler (2006): the MAE scaled by the in-sample MAE of the seasonal naive method

$$\text{MASE} = \frac{\text{MAE}}{\frac{\mathbf{1}}{n-m}\sum_{t=m+\mathbf{1}}^{n} |y_t - y_{t-m}|}$$

  - n: the length of the training sample; MASE < 1: better than the seasonal naive method in sample; MASE $= 1.2$: 20% worse

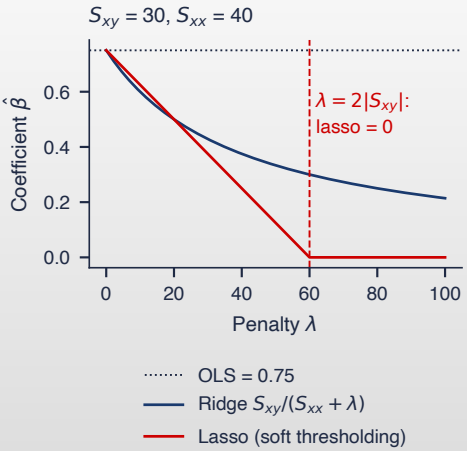
Prerequisites for Today (7/17): ridge and lasso

- **Linear model**: $\hat{y}_t = \mathbf{x}_t^\top \beta = \sum_{j=1}^{p} \beta_j x_{tj}$; ordinary least squares (OLS) minimises $\sum_t (y_t - \mathbf{x}_t^\top \beta)^2$
  - many lags and features: OLS overfits, the coefficients become large and unstable
- **Penalised least squares**: fit plus a penalty that shrinks the β_j towards 0

$$\text{ridge: } \min_\beta \sum_t (y_t - \mathbf{x}_t^\top \beta)^2 + \lambda \sum_j \beta_j^2, \qquad \text{lasso: } \min_\beta \sum_t (y_t - \mathbf{x}_t^\top \beta)^2 + \lambda \sum_j |\beta_j|$$

  - ridge Hoerl and Kennard (1970); lasso Tibshirani (1996); $\lambda \geq 0$: the strength of the penalty; $\lambda = 0$: OLS; λ large: all coefficients near 0
  - the features are **standardised** first (mean 0, standard deviation 1), so that one λ treats them equally; λ is chosen by walk-forward validation
- The lasso sets some $\hat{\beta}_j$ exactly to 0: it selects features; ridge only shrinks them

Prerequisites for Today (8/17): ridge and lasso with one feature



- One standardised feature x_t; define

$$S_{xy} = \sum_t x_t y_t, \qquad S_{xx} = \sum_t x_t^2$$

- Closed forms (minimise the two objectives in β)

$$\hat{\beta}_{OLS} = \frac{S_{xy}}{S_{xx}}, \qquad \hat{\beta}_{ridge} = \frac{S_{xy}}{S_{xx} + \lambda}$$

$$\hat{\beta}_{lasso} = \text{sign}(S_{xy}) \frac{\max(|S_{xy}| - \lambda/2,\ 0)}{S_{xx}}$$

  - **soft thresholding**: exactly 0 when $\lambda/2 \geq |S_{xy}|$; sign$(\cdot)$: $+1$ or -1

- Chart ($S_{xy} = 30$, $S_{xx} = 40$): ridge tends to 0 slowly; lasso falls linearly and is 0 from $\lambda = 60$

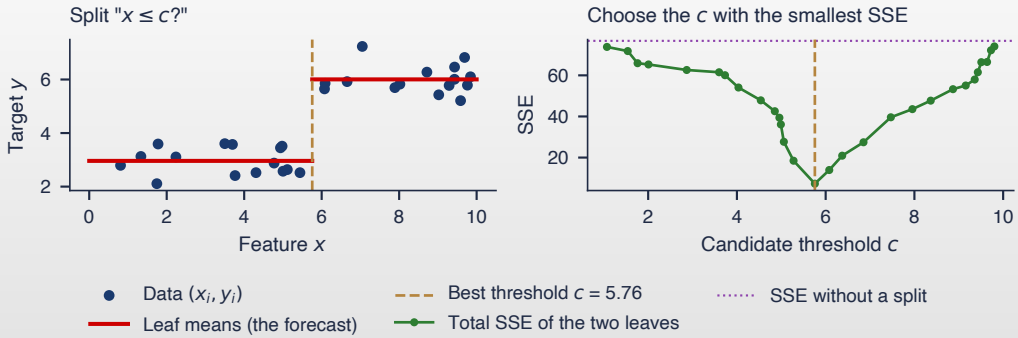
Prerequisites for Today (9/17): a regression tree

- **Regression tree**: a sequence of questions "$x_j \le c$?"; each final group (**leaf**) predicts the mean of its targets
  - x_j: one feature; c: the threshold; a deep tree has many leaves
- **One split**: for a threshold c, the left leaf $L = \{i : x_i \le c\}$ and the right leaf $R = \{i : x_i > c\}$

$$\mathrm{SSE}(c) = \sum_{i \in L} (y_i - \bar{y}_L)^2 + \sum_{i \in R} (y_i - \bar{y}_R)^2$$

  - $\bar{y}_L$, $\bar{y}_R$: the means of the targets in the two leaves; SSE: the sum of squared errors
  - the best c minimises $\mathrm{SSE}(c)$; candidates: the midpoints between consecutive sorted values of x
- A tree predicts leaf means: it cannot forecast outside the range of the training targets (no extrapolation of a trend)

Prerequisites for Today (10/17): one split in a chart



- Left: 30 simulated points; the best split puts the threshold where the level of y changes; each leaf forecasts its mean
- Right: SSE(c) for every candidate threshold; the minimum is far below the SSE without a split

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Prerequisites for Today (11/17): random forests and gradient boosting

- **Random forest** Breiman (2001): the average of B deep trees, each grown on a bootstrap sample of the rows

$$\hat{f}_{RF}(\mathbf{x}) = \frac{1}{B} \sum_{b=\mathbf{1}}^{B} \hat{f}_b(\mathbf{x})$$

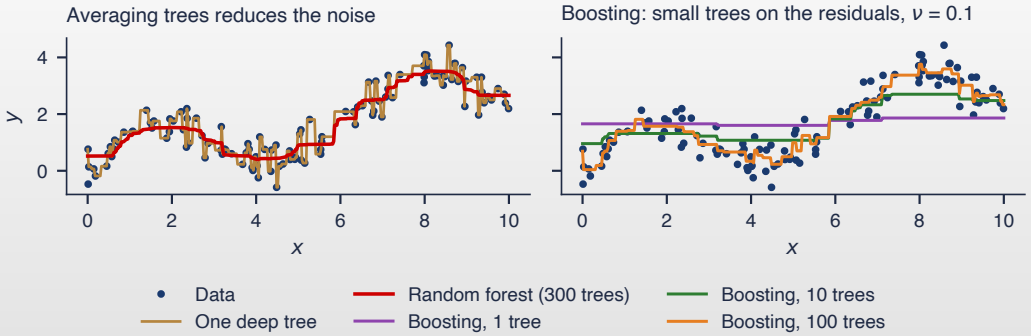
  - bootstrap sample: n rows drawn with replacement; $\hat{f}_b$: the tree b; averaging reduces the variance of single trees

- **Gradient boosting** (GB) Friedman (2001): small trees added one by one, each fitted to the current residuals

$$F_{\mathbf{0}} = \bar{y}, \qquad F_m(\mathbf{x}) = F_{m-\mathbf{1}}(\mathbf{x}) + \nu\, h_m(\mathbf{x}), \quad m = 1, \ldots, M$$

  - h_m: a small tree fitted to the residuals $y_i - F_{m-\mathbf{1}}(\mathbf{x}_i)$; $\nu \in (0, 1]$: the learning rate; M: the number of trees
  - scikit-learn: `HistGradientBoostingRegressor`; the same family as LightGBM and XGBoost

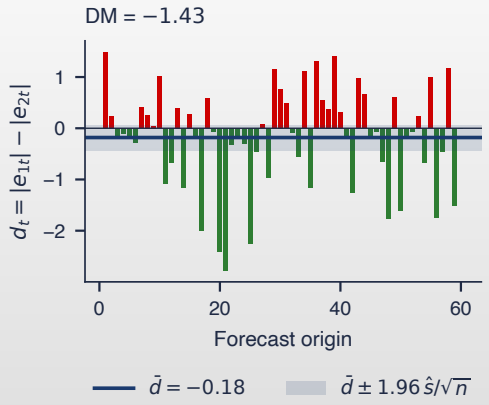
Prerequisites for Today (12/17): ensembles in a chart



- Left: one deep tree follows every noisy point; the average of 300 trees on bootstrap samples is smoother
- Right: boosting with $\nu = 0.1$; after 1 tree the fit is almost flat, after 100 trees it follows the curve

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Prerequisites for Today (13/17): the Diebold–Mariano test



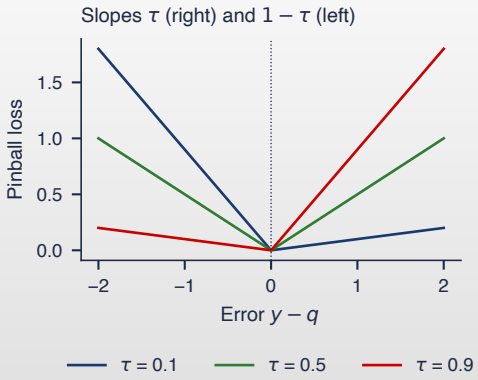
- Two forecasts of the same n targets; errors e_{1t}, e_{2t}; loss L (absolute or squared) Diebold and Mariano (1995)

$$d_t = L(e_{1t}) - L(e_{2t}), \qquad \text{DM} = \frac{\bar{d}}{\sqrt{\hat{s}^2/n}}$$

  - $\bar{d}$, $\hat{s}^2$: the mean and variance of d_t; H_0: equal accuracy, $\mathbb{E}(d_t) = 0$
- Reading the value
  - DM < 0: model 1 has the smaller loss; reject H_0 at 5% if |DM| exceeds the 97.5% quantile of $t(n-1)$
  - **HLN** correction Harvey, Leybourne and Newbold (1997) for small n and horizon h: multiply DM by $\sqrt{(n+1-2h+h(h-1)/n)/n}$
- Chart: 60 simulated origins, $\bar{d} < 0$, but $|\text{DM}| = 1.43 < 2.00$: the difference is not significant

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Prerequisites for Today (14/17): quantile forecasts and the pinball loss



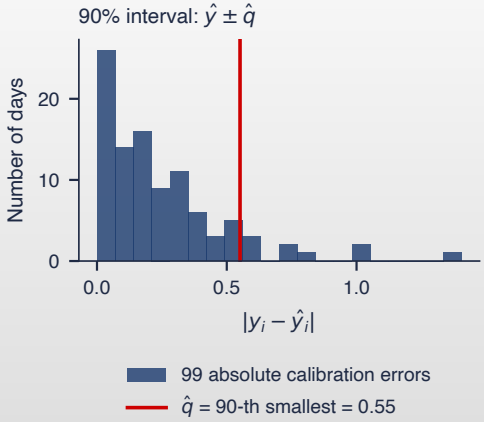
- A τ-**quantile forecast** q: a value that the outcome y should exceed with probability $1 - \tau$
  - $\tau \in (0, 1)$: the quantile level; $\tau = 0.9$: 90% of the outcomes should lie below q
- **Pinball loss** Koenker and Bassett (1978)

$$\rho_\tau(y, q) = \begin{cases} \tau\,(y - q), & y \geq q \\ (1 - \tau)(q - y), & y < q \end{cases}$$

  - for a high τ, an outcome above q costs more than one below: the loss pushes q up
  - the average pinball loss is smallest at the true τ-quantile; $\tau = 0.5$: half the absolute error (the median)

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Prerequisites for Today (15/17): split conformal intervals



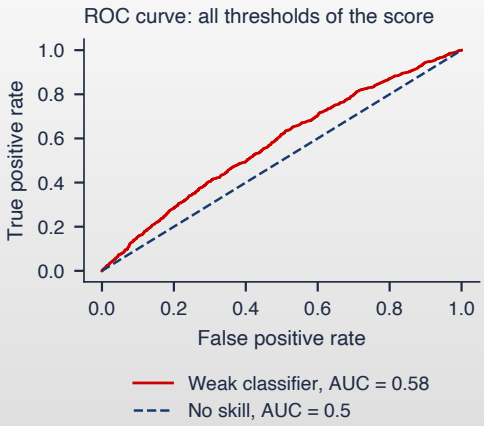
- **Split conformal** Lei et al. (2018): an interval with guaranteed coverage, for any point forecast $\hat{y}$
  - keep n **calibration** days not used in training; compute their absolute errors $|y_i - \hat{y}_i|$
- Interval with coverage $1 - \alpha$

  $$\hat{y} \pm \hat{q}, \qquad \hat{q} = \text{the } \lceil (n+1)(1-\alpha) \rceil \text{-th smallest error}$$

  - α: the allowed miscoverage ($\alpha = 0.1$: a 90% interval); $\lceil \cdot \rceil$: rounding up
  - valid if the errors are exchangeable (no change of regime between calibration and test)
- Chart: $n = 99$, $\alpha = 0.1$: $\lceil 100 \cdot 0.9 \rceil = 90$, so $\hat{q}$ is the 90th smallest error, 0.55

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Prerequisites for Today (16/17): classification of the sign



- Target $1\{r_{t+\mathbf{1}} > 0\}$: 1 if tomorrow's return is positive, else 0; **logistic regression**:

$$\Pr(r_{t+\mathbf{1}} > 0 \mid \mathbf{x}_t) = \frac{1}{1 + e^{-\mathbf{x}_t^\top \beta}}$$

  - $1\{\cdot\}$: 1 if the condition holds; predict "up" if the probability exceeds 0.5

- **Accuracy** acc: the share of correct predictions; **baseline** $\text{acc}_\mathbf{0}$: the majority class of the training data

$$z = \frac{\text{acc} - \text{acc}_\mathbf{0}}{\sqrt{\text{acc}_\mathbf{0}(1 - \text{acc}_\mathbf{0})/n}} \approx N(0,1)$$

  - n: the number of test days; $z > 1.645$: better than the baseline at 5% (one-sided)

- **AUC**: the area under the ROC curve; 0.5: no skill, 1: perfect ranking

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Prerequisites for Today (17/17): out-of-sample fit; global and local models

- **Out-of-sample** R^2: the share of the variance of the test targets explained by the forecasts

$$R^2_{oos} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y}_{train})^2}$$

  - $\bar{y}_{train}$: the mean of the training targets, the simplest forecast; $R^2_{oos} < 0$: worse than this mean
- **Local** model: trained on one series; **global** model: one model trained on many related series
  - the global model has many more rows; the series are made comparable (scaled) and a series identifier can be a feature
  - the random walk ($\hat{y}_{t+h} = y_t$) and an AR model stay the benchmarks

A1: a lag table and two strategies [Solved]

Task

- Series $y_1, \ldots, y_8$: 10, 12, 11, 13, 14, 13, 15, 16.
- 1. Build the table with the features y_{t-1}, y_{t-2} and the target y_t; how many rows does it have?
- 2. With the fitted model $\hat{y}_t = 2 + 0.6\, y_{t-1} + 0.3\, y_{t-2}$, compute the recursive forecasts $\hat{y}_9$, $\hat{y}_{10}$, $\hat{y}_{11}$.
- 3. A direct two-step model is $\hat{y}_{t+2} = 3 + 0.5\, y_t + 0.3\, y_{t-1}$: compute $\hat{y}_{10}$ and the number of rows of its table.
- Report: the table, three recursive forecasts, one direct forecast, two row counts.

Solution

t	y_{t-1}	y_{t-2}	y_t
3	12	10	11
4	11	12	13
5	13	11	14
6	14	13	13
7	13	14	15
8	15	13	16

- 1. 6 rows ($t = 3, \ldots, 8$)
- 2. $\hat{y}_9 = 2 + 0.6 \cdot 16 + 0.3 \cdot 15 = 16.10$; $\hat{y}_{10} = 2 + 0.6 \cdot 16.10 + 0.3 \cdot 16 = 16.46$; $\hat{y}_{11} = 16.71$
- 3. $\hat{y}_{10} = 3 + 0.5 \cdot 16 + 0.3 \cdot 15 = 15.50$; 5 rows (origins $t = 2, \ldots, 6$)

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A2: identifying data leakage [Proposed]

Solution discussed in class

Task

- Daily series 10, 12, 11, 13, 14; the target is the last value, 14. Model: A1.
- 1. Compute the correct 3-day mean feature for this target, and the mean that a careless `rolling(3).mean()` without `shift(1)` would give.
- 2. For each choice, say whether it leaks: (a) a scaler fitted on 2022–2026 before the split; (b) holiday dummies of the target day; (c) the temperature measured on the target day; (d) 5-fold shuffled CV for a 21-day target.
- Report: two means and four verdicts with one reason each.

A3: one split of a regression tree [Solved]

Task

- Six days: x = load on the same weekday last week, y = load today (GW): $(4.8, 5.0)$, $(5.0, 5.2)$, $(5.6, 5.9)$, $(6.0, 6.0)$, $(6.4, 6.6)$, $(6.9, 7.0)$.
- 1. List the five candidate thresholds.
- 2. For each, compute the two leaf means and the total SSE.
- 3. Choose the split and give the forecast for $x = 6.2$.
- Report: a table of five rows, the chosen threshold, one forecast.

Solution

threshold	left mean	right mean	SSE
4.9	5.00	6.14	1.912
5.3	5.10	6.38	0.827
5.8	5.37	6.53	0.953
6.2	5.53	6.80	0.828
6.7	5.74	7.00	1.672

- Without a split: mean 5.95, SSE 2.995
- Best: $x \leq 5.3$, SSE 0.827; leaves 5.10 and 6.38 GW
- $x = 6.2 > 5.3$: forecast 6.38 GW

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A4: ridge and lasso in one dimension [Proposed]

Solution discussed in class

Task

- One standardised feature with $S_{xy} = 40$ and $S_{xx} = 50$. Model: A3 and the formulas of Prerequisites for Today (8/17).
- 1. Compute the OLS coefficient.
- 2. Compute the ridge coefficient for $\lambda = 10$ and $\lambda = 50$.
- 3. Compute the lasso coefficient for $\lambda = 20$ and $\lambda = 100$.
- 4. Explain in one sentence which method can drop a lag from the model.
- Report: five coefficients and one sentence.

A5: a conformal interval and the pinball loss [Solved]

Task

- A model forecasts tomorrow's load $\hat{y} = 6.10$ GW. On 9 calibration days its absolute errors were 0.12, 0.30, 0.05, 0.22, 0.41, 0.18, 0.09, 0.27, 0.35 GW.
- 1. Build the 80% split conformal interval.
- 2. A 90% quantile forecast is $q = 6.4$ GW: compute its pinball loss if $y = 6.0$ and if $y = 6.6$.
- Report: k, $\hat{q}$, the interval, two losses.

Solution

- 1. Sorted: 0.05, 0.09, 0.12, 0.18, 0.22, 0.27, 0.30, 0.35, 0.41; $k = \lceil 10 \cdot 0.8 \rceil = 8$; $\hat{q} = 0.35$
- interval $6.10 \pm 0.35 = [5.75, 6.45]$ GW
- 2. $y = 6.0 < q$: $(1 - 0.9)(6.4 - 6.0) = 0.04$; $y = 6.6 > q$: $0.9 \cdot 0.2 = 0.18$
- an outcome above a high quantile costs nine times more per GW: the loss pushes q up until only 10% of outcomes exceed it

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A6: MASE and a Diebold–Mariano statistic [Proposed]

Solution discussed in class

Task

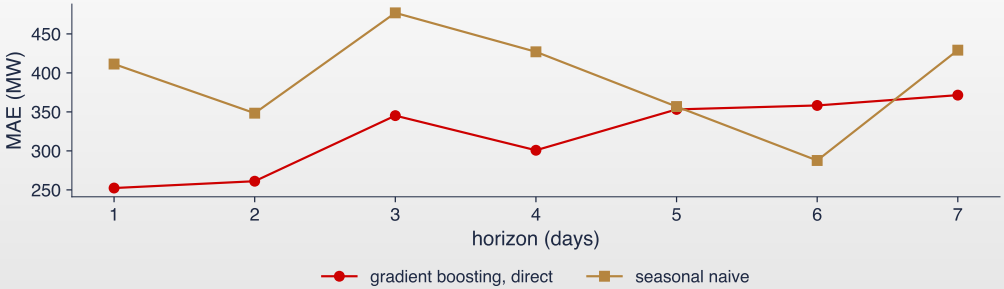
- Average absolute errors (GW) of two models at 6 origins: A = 0.30, 0.25, 0.40, 0.28, 0.35, 0.22; B = 0.32, 0.30, 0.38, 0.35, 0.41, 0.30; MASE scale 0.31 GW. Model: A5, Prerequisites for Today (6/17) and Prerequisites for Today (13/17).
- 1. Compute the MAE and the MASE of each model.
- 2. Compute $d_t = A_t - B_t$, its mean $\bar{d}$ and its standard deviation s, then $DM = \bar{d}/(s/\sqrt{6})$.
- 3. Compare $|DM|$ with the 5% critical value of $t(5)$ and conclude.
- Report: four accuracy numbers, $\bar{d}$, DM and a decision.

B1: gradient boosting for the electricity load [Solved]

- **Question**: does a gradient-boosting model forecast next week's daily load better than "the same weekday last week"?
- **Inputs**: daily mean load of Romania (GW), 2022–2026; weekly origins from 5 January to 22 June 2026; horizons 1–7 days
- **Tasks**
  1. Build the direct tables (14 lags, 7- and 28-day means, the same weekday, calendar of the target day) with `direct_frame`.
  2. At each origin, train one `HistGradientBoostingRegressor` per horizon on the data before the origin and forecast the next 7 days.
  3. Compute the MAE, RMSE and MASE of GB and of the seasonal naive forecast.
  4. Run the Diebold–Mariano test on the origin-average absolute errors.
  5. Interpretation: at which horizons does GB gain most, and why?
- **Report**: six accuracy numbers, a DM statistic with its p-value, one sentence
- Notebook: section B1

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B1: solution [Solved]



- 25 origins, 7 horizons; MAE: GB 320 MW, seasonal naive 391 MW; RMSE 409 and 521 MW; MASE 1.04 and 1.27 (scale 308 MW)
- DM (with the Harvey–Leybourne–Newbold small-sample correction, HLN) $= -2.63$, $p = 0.015$; the worst origin for GB: 5 January 2026 (MAE 576 MW)
- Interpretation: one day ahead GB has 252 MW against 411 MW: it uses yesterday's level, which the seasonal naive ignores; at 7 days (371 and 429 MW) both rely on last week's pattern

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B2: recursive, direct and the value of holidays [Proposed]

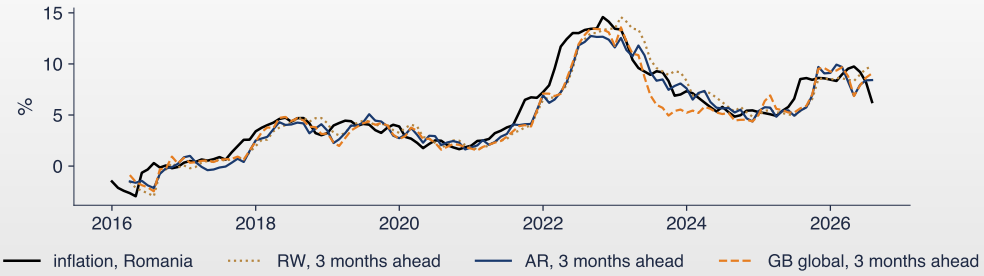
- **Question**: does the recursive strategy forecast as well as the direct one, and how much do the holiday features help?
- **Inputs**: the data and origins of B1; model: B1
- **Tasks**
  1. Repeat B1 with the recursive strategy (`forecast_recursive`).
  2. Repeat the direct model without the three holiday features (easter, xmas, holiday).
  3. Report the MAE of the three models on all days, on the public-holiday days and on the other days.
  4. Interpretation: which feature group explains most of the difference, and on which days?
- **Report**: nine MAE values and one sentence
- Notebook: section B2

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B3: Romanian inflation three months ahead [Solved]

- **Question**: can machine learning forecast Romanian inflation three months ahead better than the random walk and a linear AR model?
- **Inputs**: 12-month HICP inflation of 27 EU countries (Eurostat), monthly, 2005–2026
- **Tasks**
  1. Build the features of each country (inflation lags, monthly changes, calendar month) and the target $\pi_{t+3} - \pi_t$ with `inflation_frame`.
  2. Every January from 2016, refit AR, lasso, random forest and GB on Romania only, and a global GB on all countries; forecast every month of the year.
  3. Compute the RMSE of each model and its ratio to the random walk.
  4. Run the DM test for AR against RW and for the global against the local GB.
  5. Interpretation: what happened to all models in 2021–2023?
- **Report**: six RMSEs, two DM tests and one sentence
- Notebook: section B3

B3: solution [Solved]



- 125 origins, January 2016–May 2026; RMSE (pp): RW 1.41, AR 1.28 (0.90), lasso 1.45 (1.02), RF 1.49 (1.06), GB local 1.64 (1.16), GB global 1.37 (0.97)
- DM: AR against RW -1.33 ($p = 0.186$); GB global against GB local -1.60 ($p = 0.113$)
- Interpretation: from mid-2021 to mid-2023 every RMSE jumps (RW 2.27, AR 2.02, GB global 2.09 pp): the energy shock was not in the past of inflation; pooling 27 countries helps the trees, but no model is significantly better than AR

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B4: the sign of BET returns [Proposed]

- **Question**: can a classifier predict whether the BET rises tomorrow better than "it always rises"?
- **Inputs**: daily BET log returns since 2000; test years 2015–2026; model: B3 (yearly refits)
- **Tasks**
  1. Build the features of `sign_frame` (five lagged returns, sums over 5, 21, 63 days, volatility over 21 and 63 days) and the target $1\{r_{t+1} > 0\}$.
  2. Each January, fit a logistic regression and a gradient-boosting classifier on all earlier days; the baseline predicts the majority class of the training data.
  3. Report the accuracy of the three, the z statistic against the baseline and the AUC (the probability that a random up day gets a higher score than a random down day; 0.5: no skill).
  4. Interpretation: what does the result say about the weak-form efficiency of the Bucharest market?
- **Report**: three accuracies, two z statistics, two AUCs and one sentence
- Notebook: section B4

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C1: a global model for European electricity load [Proposed]

- **Idea**: ENTSO-E publishes the hourly load of every European country; one global GB model trained on 20 countries may forecast Romania better than a local one
  - features: lags scaled by each country's mean, weekday, national holidays, a country identifier
- Questions to answer
  - 1. Does the global model beat the local GB and DHR on the 26 origins of Chapter 4?
  - 2. Does it forecast Orthodox Easter better, although most countries celebrate Western Easter?
  - 3. Which countries help Romania most (leave-one-country-out)?
- A possible team project: data, code, DM tests and a two-page report; model: B1, B3

C2: audit an AI answer [Proposed]

- A student asked an AI assistant whether machine learning can forecast markets and electricity load. The answer:
- (a) A random forest with 5-fold cross-validation explains 36.8% of the next month's S&P 500 return, so the market is predictable.
- (b) A logistic model predicts the daily sign with 53.8% accuracy, well above 50%: a profitable signal.
- (c) For the load, GB has MASE 0.97, so it is 3% more accurate than DHR.
- (d) Standardise the whole series first, then split it into training and test.
- (e) An LSTM always beats ARIMA because it has long memory.
- (f) Calendar features of the target day, such as holidays, are allowed because they are known in advance.
- **Tasks**
  1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the lecture notebook (sections 2, 6, 9).
- **Report**: a list of six verdicts with one line of justification each

Key takeaways

- A forecast is a prediction from features known at the origin: lags, past rolling means, calendar
- Recursive and direct strategies: compare them on the same origins
- Leakage: any value from after the origin, including through scaling or shuffled folds
- GB beats the seasonal naive forecast for the load; nothing beats "always up" for the sign of BET returns
- An AI answer is a draft: check the validation scheme and the benchmark behind every claimed accuracy

After the seminar

- Lecture 9 develops each topic of today
  - features and strategies, walk-forward validation, ridge and lasso, trees and boosting, neural networks and the LSTM, prediction intervals, global models, the M4 and M5 competitions
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: a global model for European electricity load
- Reading: Huang and Petukhina (2022), Ch. 10; Hyndman and Athanasopoulos (2021), Sec. 12.4; Montero-Manso and Hyndman (2021)

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